

Theory of Computation (TOC) – Revision Definitions **1. Set**

A set is a well-defined collection of distinct objects called elements.

2. Subset

A set A is called a subset of set B if every element of A is also an element of B.

3. Complement

The complement of a set A consists of all elements in the universal set U that are not in A.

4. One-to-One Function (Injective)

A function $f:A \rightarrow B$ is one-to-one if distinct elements of A have distinct images in B.

5. Onto Function (Surjective)

A function $f:A \rightarrow B$ is onto if every element of B is the image of at least one element of A.

6. Bijective Function

A function that is both one-to-one and onto is called a bijection.

7. Inverse Function

If a function f is bijective, then there exists a function f^{-1} called the inverse function.

8. Partial Function

A function that is defined only for some elements of its domain is called a partial function.

9. Total Function

A function that is defined for every element of its domain is called a total function.

10. Constant Function

A function that maps every element of its domain to the same value is called a constant function.

11. Recursive Function

A function defined in terms of itself together with one or more base cases is called a recursive function.

12. Primitive Recursive Function

A function obtained from initial functions using finite applications of composition and primitive recursion is called a primitive recursive function.

13. Finite Automata (FA)

A finite automaton is a mathematical model consisting of a finite number of states used to recognize regular languages.

14. NFA

A Non-deterministic Finite Automaton is a finite automaton in which more than one transition may exist for a state and input symbol.

15. ϵ -Closure(q)

The ϵ -closure of a state q is the set of all states reachable from q using zero or more ϵ -transitions.

16. Regular Language

A language accepted by a finite automaton is called a regular language.

17. Regular Expression

A notation used to describe regular languages is called a regular expression.

18. Regular Grammar

A grammar in which every production is either $A \rightarrow aB$ or $A \rightarrow a$ is called a regular grammar.

19. Context Free Grammar (CFG)

A grammar in which every production has a single variable on the left side is called a context-free grammar.

20. Context Free Language (CFL)

A language generated by a context-free grammar is called a context-free language.

21. Ambiguous Grammar

A CFG is ambiguous if there exists at least one string having more than one parse tree.

22. Chomsky Hierarchy

Classification of grammars into Type 0, Type 1, Type 2 and Type 3 grammars.

23. Chomsky Normal Form (CNF)

A CFG is in CNF if every production is of the form $A \rightarrow BC$ or $A \rightarrow a$.

24. Pushdown Automata (PDA)

A PDA is a finite automaton equipped with an additional memory structure called a stack.

25. Deterministic PDA (DPDA)

A PDA having at most one possible move for every state and input combination.

26. Acceptance by PDA

Acceptance by Final State and Acceptance by Empty Stack.

27. Operations on PDA

Push, Pop and Replace.

28. Pumping Lemma for Regular Languages

Used to prove that a language is not regular.

29. Pumping Lemma for CFL

Used to prove that a language is not context-free.

30. Recursive Language

A language accepted by a Turing Machine that always halts.

31. Recursively Enumerable Language

Accepted by a Turing Machine that may not halt on rejected strings.

32. Decidable Language

A language is decidable if a Turing Machine always halts and decides membership correctly.

33. Halting Problem

The problem of determining whether a Turing Machine halts on a given input. It is undecidable.

34. Ogden's Lemma

A stronger version of the pumping lemma for context-free languages.

35. Complement of CFL

The class of CFLs is not closed under complement.

36. Intersection of CFLs

The class of CFLs is not closed under intersection.

37. Rounded (Bounded) Quantification

A quantifier whose range is restricted to a finite set of values.

38. Moore Machine

A finite-state machine whose output depends only on the current state.

39. Mealy Machine

A finite-state machine whose output depends on both current state and input symbol.

40. Class P

Decision problems solvable by a deterministic Turing Machine in polynomial time.

41. Class NP

Decision problems whose solutions can be verified in polynomial time.

42. NP-Complete Problem

A problem that belongs to NP and every NP problem can be reduced to it in polynomial time.